STEP BY STEP PROCEDURE FOR KUBERNETES

1. Create a very small web app (Python Flask).
2. Create a Dockerfile and build the image.
3. Push the image to Docker Hub (or use local image with minikube).
4. Create Kubernetes Deployment and Service YAML files.
5. Deploy using kubectl and verify.
6. Optional: simpledeploy script.

1. deployment.yaml:

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: registration-app-deployment
  labels:
    app: registration-app
spec:
  replicas: 2  # number of pod copies
  selector:
    matchLabels:
      app: registration-app
  template:
    metadata:
      labels:
        app: registration-app
    spec:
      containers:
      - name: registration-app-container
        image: renuka1163/registration-app:latest
        imagePullPolicy: IfNotPresent
        ports:
        - containerPort: 8080  # change this if your app runs on another port
```

2. service.yaml:

```
apiVersion: v1
kind: Service
metadata:
  name: registration-app-service
spec:
  selector:
    app: registration-app
```

```
  ports:
    - protocol: TCP
      port: 80        # external port
      targetPort: 8080 # internal port (same as containerPort above)
  type: NodePort
```

STEP-BY-STEP PROCEDURE

Prerequisites: - Docker installed and running - kubectl installed - minikube or kind (local Kubernetes cluster) - Docker Hub account (for pushing image) or knowledge of using local images with minikube

1) Create project folder and files:

mkdir simple-k8s-app
cd simple-k8s-app

Create

app.py
requirements.txt
Dockerfile
deployment.yaml
service.yaml

2) Test app locally:

pip install -r requirements.txt python app.py Visit
http://localhost:8080 to confirm "Hello from Kubernetes!"

3) Build Docker image:

docker build -t yourdockerhubusername/simpleapp:latest .

4) Push image to Docker Hub:

docker login docker push
yourdockerhubusername/simpleapp:latest

ALTERNATIVE (minikube local images): minikube start eval $(minikube docker-env)docker build -t simpleapp:latest (Use image name in deployment as simpleapp:latest)

5) Deploy to Kubernetes:

kubectl apply -f deployment.yaml
kubectl apply -f service.yaml

6) Verify:

kubectl get pods
kubectl get deployments
kubectl get svc

If not working on port number use to following command:
kubectl port-forward svc/registration-app-service 8095:80
If using minikube, run: minikube service simpleapp-service